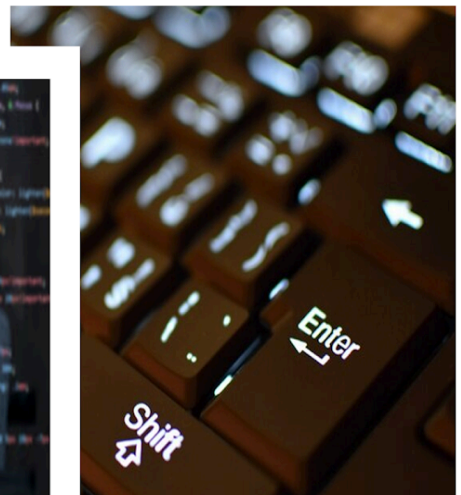


Computer Awareness

STUDY NOTES

Social Media and Artificial Intelligence



Social Media and Artificial Intelligence

1. Introduction to Social Media

- Social media refers to digital platforms that allow users to create, share, and exchange content in virtual communities.
- **Examples:** Facebook, Twitter, Instagram, LinkedIn, TikTok.

2. Importance of Social Media

- **Communication:** Facilitates instant communication and connection.
- **Information Dissemination:** A vital tool for sharing news and information quickly.
- **Marketing and Business:** Used for brand promotion, customer engagement, and market research.

3. Impact of Social Media

- **Positive Effects:**
 - Networking opportunities.
 - Access to a vast amount of information.
 - Platform for activism and social movements.
- **Negative Effects:**
 - Cyberbullying and harassment.
 - Spread of misinformation.
 - Addiction and mental health issues.

4. Introduction to Artificial Intelligence (AI)

- AI refers to the simulation of human intelligence in machines that are programmed to think and learn like humans.
- **Types of AI:**
 - **Narrow AI:** Specialized in one task (e.g., virtual assistants, recommendation systems).
 - **General AI:** Hypothetical AI that possesses the ability to perform any intellectual task that a human can do.

5. Importance of AI

- **Efficiency:** Automates repetitive tasks and processes.
- **Data Analysis:** Analyzes vast amounts of data for insights and predictions.
- **Enhanced User Experience:** Personalizes services and recommendations.

6. Applications of AI

- **Healthcare:** Diagnostics, personalized medicine, robotic surgery.
- **Finance:** Fraud detection, algorithmic trading, credit scoring.
- **Transportation:** Autonomous vehicles, traffic management systems.

7. Intersection of Social Media and AI

- **Content Recommendation:** AI algorithms analyze user behavior to suggest content on social media platforms.
- **Sentiment Analysis:** AI tools analyze social media posts to gauge public opinion on various issues.
- **Chatbots:** AI-driven bots provide customer service and engagement on social media.

8. Challenges and Concerns

- **Privacy Issues:** Data collection and surveillance concerns.
- **Manipulation of Information:** Use of AI for targeted misinformation campaigns.
- **Ethical Considerations:** Bias in AI algorithms affecting decision-making.

9. Current Trends

- **Influencer Marketing:** The rise of influencers and their impact on consumer behavior.
- **AI in Content Creation:** Tools that generate articles, music, and art using AI.
- **Augmented Reality (AR) and Virtual Reality (VR):** Enhancements in social media engagement.

Type of Chatbot	Example	Description
Customer Service	Zendesk, Cathay Pacific, Intercom, Botsify, Amtrak etc	Provides automated responses to common customer inquiries and directs complex questions to human agents.
E-commerce	Sephora, Domino's, Octane AI, Tidio etc	Assists shoppers in finding products, tracking orders, and managing their accounts on e-commerce platforms.
Personal Assistant	Google Assistant, Siri, Alexa, Bixby etc.	Performs tasks like setting reminders and controlling smart home devices via voice or text commands.

Banking	Erica (Bank of America), Amy From Hsbc, Kasisto etc	Offers financial assistance, including balance checks, money transfers, and budgeting advice through AI interaction.
Health and Wellness	Woebot, Florence, Aiden, JIVA etc	Provides mental health support using cognitive behavioral therapy techniques and mood tracking.
Educational	Duolingo Chatbot	Helps users practice languages by simulating conversations and giving instant feedback.
Travel	KAYAK Chatbot	Assists users in searching for travel deals and managing their itineraries.
Social Media	Messenger Bots (Facebook)	Engages users with automated responses and interactive content within the Facebook Messenger app.
Restaurant	Domino's Pizza Chatbot	Allows customers to order pizza and customize their meals through chat.
Event	Eventbrite Chatbot	Helps users find events and purchase tickets while answering related FAQs.
Industrial Robot	ABB IRB 6700	Used in manufacturing for tasks like welding, painting, and assembly.
Service Robot	Roomba	A robotic vacuum cleaner that autonomously cleans floors using sensors and algorithms.
Humanoid Robot	Sophia	A social humanoid robot designed to interact with humans and display emotions.
Surgical Robot	da Vinci Surgical System	Assists surgeons in performing minimally invasive surgeries with precision.
Agricultural Robot	Harvest Robotics CROO	Automated robot designed for harvesting strawberries and other crops.

Military Robot	PackBot	Used by military and first responders for bomb disposal and reconnaissance.
Educational Robot	LEGO Mindstorms	A programmable robot kit used for teaching programming and robotics concepts.
Autonomous Vehicle	Waymo	Self-driving car technology designed for safe and efficient transportation.
Drones	DJI Phantom	Unmanned aerial vehicles used for photography, surveying, and delivery services.
Telepresence Robot	Double Robotics	Allows remote users to interact and communicate in different locations via video conferencing.

Website	Description
Google	Uses AI for search algorithms, personalized recommendations, and voice recognition through Google Assistant.
OpenAI	Offers AI research and applications, including the ChatGPT model for conversational AI and other tools for developers.
IBM Watson	Provides AI-powered solutions for data analysis, natural language processing, and machine learning applications.
Amazon Alexa	A voice-controlled virtual assistant that uses AI for voice recognition, smart home integration, and information retrieval.
Microsoft Azure	Offers a suite of AI services, including machine learning, cognitive services, and bot services for developers and businesses.
Hugging Face	A platform for sharing and deploying natural language processing models and tools, focusing on AI for conversational applications.

ChatGPT (by OpenAI)	A conversational AI platform that allows users to engage with an AI model for various applications, from customer service to content generation.
Replika	An AI chatbot designed for companionship and conversation, utilizing natural language processing to learn from user interactions.
Zest AI	Focuses on using AI for credit underwriting and financial services, providing tools to assess risk and improve lending decisions.
Cortana	Microsoft's virtual assistant that leverages AI for task management, reminders, and integration with Microsoft Office applications.